

HAVRE ST PIERRE, QC, CanadaWMO: **715840**Lat: **50.282N**Lon: **63.611W**Elev: **124**StdP: **14.63**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-10.8	-26.7	1.2	-14.3	-22.0	1.6	-9.4	35.2	20.3	30.6	20.4	8.1	280	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	73.1	62.9	70.6	61.6	68.2	60.3	65.4	70.2	63.7	67.9	62.1	66.1	10.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.3	67.4	61.9	83.2	65.9	60.3	78.4	64.3	30.4	70.1	29.1	68.1	28.0	66.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.4	23.5	20.8	-23.8	79.3	4.5	2.2	-27.1	80.8	-29.7	82.1	-32.2	83.3	-35.5	84.9		
(5)			-24.2	69.2	4.4	1.9	-27.4	70.5	-30.0	71.6	-32.5	72.7	-35.7	74.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	9.2	10.7	19.9	32.0	43.2	53.2	59.7	58.9	50.7	40.8	29.6	16.9		
	DBStd	19.24	11.52	11.43	10.38	6.27	5.39	5.23	3.94	4.17	5.53	5.64	8.34	10.89		
	HDD50	6069	1264	1100	933	541	221	26	0	1	55	288	613	1027		
	HDD65	10764	1729	1520	1398	991	675	355	171	193	428	749	1063	1492		
	CDD50	788	0	0	0	0	11	121	300	276	77	3	0	0		
	CDD65	9	0	0	0	0	0	0	5	3	1	0	0	0		
	CDH74	46	0	0	0	0	0	3	27	14	2	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	9.6	10.7	10.5	10.8	10.4	9.2	9.3	8.3	8.0	8.8	9.4	9.7	10.7		
(15) Precipitation	PrecAvg	41.7	3.0	2.3	3.1	2.5	3.1	3.4	4.3	3.7	4.1	4.8	4.3	3.8		
	PrecMax	50.9	5.0	3.6	5.8	5.3	5.2	5.7	6.5	6.7	6.5	8.6	7.9	6.5		
	PrecMin	36.4	0.9	0.9	1.7	0.9	1.1	1.3	1.7	2.0	1.9	1.7	2.2	2.4		
	PrecStd	4.2	1.3	0.7	1.2	1.1	1.3	1.3	1.3	1.2	1.2	1.6	1.3	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	39.6	42.7	52.6	67.3	73.8	77.2	75.8	70.5	59.4	51.9	42.5		
		MCWB	37.0	37.2	37.0	42.7	56.0	61.7	66.1	66.1	61.5	53.8	50.6	41.3		
	2%	DB	34.2	36.0	38.8	47.1	60.9	69.7	73.6	72.4	65.8	55.4	46.8	38.7		
		MCWB	33.1	33.4	34.2	40.2	50.9	58.9	63.3	63.3	58.9	50.4	45.3	37.2		
	5%	DB	31.4	32.4	36.1	43.6	56.9	66.6	71.1	70.0	62.8	52.2	43.5	35.7		
		MCWB	30.5	30.8	32.5	37.9	48.5	56.9	62.0	62.2	56.9	48.7	41.6	34.2		
	10%	DB	27.6	28.3	33.8	40.8	53.6	63.6	68.5	67.6	60.0	49.9	41.0	33.0		
		MCWB	26.4	26.3	31.1	36.2	46.3	55.3	60.9	60.8	54.9	46.7	39.3	31.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	38.4	38.0	44.7	56.8	65.0	68.3	67.8	64.0	55.8	51.1	41.3		
		MCDB	38.6	39.3	41.1	49.6	63.7	70.4	74.5	73.0	67.8	58.3	51.4	41.7		
	2%	WB	33.3	33.6	35.3	41.4	52.4	61.2	65.6	65.4	60.6	52.0	45.7	37.4		
		MCDB	34.3	35.1	37.4	45.9	58.4	67.0	70.7	69.9	63.9	53.8	46.4	38.3		
	5%	WB	30.6	30.7	33.2	38.9	49.6	58.8	63.9	63.7	58.2	49.6	42.1	34.5		
		MCDB	31.4	32.5	35.2	42.6	55.1	64.4	68.4	67.5	61.0	51.4	43.2	35.4		
	10%	WB	26.2	26.6	31.5	36.9	47.4	56.5	62.3	62.2	56.0	47.3	39.5	32.0		
		MCDB	27.5	28.1	33.7	39.9	52.5	61.8	66.8	65.7	59.0	49.2	40.7	33.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.7	16.7	13.4	15.8	16.6	16.0	16.3	16.0	13.5	12.4	14.5		
		MCDBR	17.9	13.8	14.5	17.4	22.6	22.0	20.4	19.8	19.3	16.5	12.8	12.4		
		MCWBR	17.9	13.1	11.7	12.2	14.0	12.5	12.0	11.9	12.7	12.5	11.9	12.2		
	5% WB	MCDBR	17.9	14.4	13.1	14.9	19.8	18.6	17.0	16.5	16.6	14.4	12.3	13.0		
		MCWBR	18.0	14.2	11.4	11.3	13.1	11.9	11.0	11.0	12.2	12.2	12.2	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.295	0.334	0.363	0.381	0.415	0.379	0.353	0.331	0.291	0.271		
	taud		2.352	2.330	2.327	2.353	2.403	2.379	2.335	2.440	2.484	2.509	2.509	2.365		
	Ebn at Noon		249	279	290	289	284	278	266	271	268	253	240	228		
	Edh at Noon		20	27	32	36	36	36	38	32	28	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		331	623	1053	1427	1651	1757	1677	1439	1087	667	348	260		
	RadStd		42	58	106	127	97	167	155	64	69	45	32	25		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	+2.86	N/A	N/A	+0.76	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	+2.23	N/A	+0.59	+0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page